

ShopFlow AWS Project

CloudWatch Design

Logs - Metrics - Alarms - Dashboard

Module 16 of 18

Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon CloudWatch is the centralised observability platform for ShopFlow. It collects logs from EC2, ECS, Lambda, and RDS; evaluates metric alarms; and displays everything in a unified dashboard. All AWS services emit metrics automatically -- no agent needed for ALB, RDS, SQS, or ElastiCache metrics.

2. Log Groups

Log Group	Source	Retention
/shopverse/dev/application	EC2 CloudWatch Agent (Spring Boot logs)	30 days
/ecs/shopflow-shopverse-dev	ECS awslogs driver (container stdout/stderr)	30 days
/aws/rds/instance/.../slowquery	RDS slow query log	7 days
/aws/lambda/shopflow-email-notifier-dev	Lambda function logs	30 days
/aws/lambda/shopflow-order-processor-dev	Lambda function logs	30 days
/aws/codebuild/shopflow-build-dev	CodeBuild build output	30 days

3. Alarms

Alarm Name	Metric	Threshold	Action
shopflow-alb-5xx-errors	ALB HTTPCode_Target_5XX_Count	>10 in 5 min	SNS email
shopflow-alb-high-latency	ALB TargetResponseTime	>3s avg (2 periods)	SNS email
shopflow-rds-high-cpu	RDS CPUUtilization	>80% (2 periods)	SNS email
shopflow-rds-low-storage	RDS FreeStorageSpace	<2 GB	SNS email
shopflow-notification-dlq	SQS ApproximateNumberOfMessagesVisible	>0	SNS email

4. Dashboard -- shopflow-overview-dev

Widget	Metrics Shown
ALB Traffic	Request count (total) + HTTP 5xx error count -- 1-minute resolution
ALB Latency	Average response time + p99 response time -- 1-minute resolution
RDS	CPU utilization (%) + active DB connections -- 1-minute resolution
SQS Queue Depth	Messages visible in notification queue, DLQ, and order events queue

5. Logs Insights Sample Queries

Use Case	Query
Find all ERRORS	fields @timestamp, @message filter @message like /ERROR/ sort @timestamp desc
Slow requests	fields @message filter @message like /execution time/ sort @timestamp desc
Request count by status	fields @message stats count() by httpStatus

6. Alarm States

State	Meaning	Email Sent?
OK	Metric is below threshold	Yes (when transitioning from ALARM to OK)
ALARM	Metric crossed threshold	Yes (immediately on transition)
INSUFFICIENT_DATA	Not enough data to evaluate	No